

2019-2020-2 概率 C 参考答案 (A 卷)

一、1. $\frac{1}{3}$ 2. $\frac{30}{100}$ 3. $\frac{0.8}{1}$ 4. $\frac{5}{36}$ 5. $t(n-1)$

二、1. D 2. C 3. B 4. B 5. A

三、(10 分)

解： 设 B 表示“取得的产品是次品”， A_1, A_2, A_3 分别表示产品是由甲厂、乙厂、丙厂生产的，

(1) 由全概率公式，得

$$P(B) = \sum_{i=1}^3 P(A_i)P(B|A_i) = \frac{6}{20} \times \frac{20}{100} + \frac{4}{20} \times \frac{15}{100} + \frac{10}{20} \times \frac{10}{100} = 0.14 \quad \dots\dots\dots (5 \text{ 分})$$

$$(2) \text{ 由贝叶斯公式, 得 } P(A_2|B) = \frac{\frac{4}{20} \times \frac{15}{100}}{\frac{14}{100}} = \frac{3}{14} \quad \dots\dots\dots (5 \text{ 分})$$

四、(12 分)

解： (1) 由于 $1 = \int_{-\infty}^{+\infty} f(x)dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} A \cos x dx = 2A$, 故 $A = \frac{1}{2}$. $\dots\dots\dots (4 \text{ 分})$

$$(2) P(0 < X < \frac{\pi}{4}) = \int_0^{\frac{\pi}{4}} \frac{1}{2} \cos x dx = \frac{\sqrt{2}}{4}. \quad \dots\dots\dots (4 \text{ 分})$$

$$(3) F_Y(y) = P(Y \leq y) = P(2X \leq y) = P(X \leq \frac{y}{2}) = F_X(\frac{y}{2}),$$

$$f_Y(y) = [F_Y(y)]' = \frac{1}{2} f_X(\frac{y}{2}) = \begin{cases} \frac{1}{4} \cos \frac{y}{2}, & |y| \leq \pi, \\ 0, & |y| > \pi. \end{cases} \quad \dots\dots\dots (4 \text{ 分})$$

五、(16 分)

解： (1)

X	0	1
p	0.6	0.4

Y	-1	0	1
p	0.3	0.3	0.4

$\dots\dots\dots (4 \text{ 分})$

Z=X+Y	-1	0	1	2
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(2)	p	0.2	0.2	0.5	0.1
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..... (4分)

(3)	Z=XY	-1	0	1
	p	0.1	0.8	0.1

..... (4分)

$$(4) EX = 0 \times 0.6 + 1 \times 0.4 = 0.4$$

$$EY = (-1) \times 0.3 + 0 \times 0.3 + 1 \times 0.4 = 0.1$$

$$E(XY) = (-1) \times 0.1 + 0 \times 0.8 + 1 \times 0.1 = 0$$

$$\text{cov}(X, Y) = E(XY) - EXEY = 0 - 0.4 \times 0.1 = -0.04 \quad \text{..... (4分)}$$

六、(20分)

$$\text{解: (1) 由概率密度性质得 } A \int_0^{+\infty} dx \int_0^{+\infty} e^{-(x+y)} dy = 1, A=1 \quad \text{..... (5分)}$$

$$(2) P(X+Y < 1) = \iint_{x+y < 1} f(x, y) dx dy = \int_0^1 dx \int_0^{1-x} e^{-(x+y)} dy = 1 - \frac{2}{e} \quad \text{..... (5分)}$$

$$(3) f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_0^{+\infty} e^{-(x+y)} dy = e^{-x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_0^{+\infty} e^{-(x+y)} dx = e^{-y}, & y > 0 \\ 0, & y \leq 0 \end{cases} \quad \text{..... (5分)}$$

$$(4) \text{独立, 由 (3) 知 } f(x, y) = f_X(x) f_Y(y) \text{ 所以独立} \quad \text{..... (5分)}$$

七、(12分)

解: (1) 矩估计:

$$EX = \int_0^1 x(\theta+1)x^\theta dx = \frac{\theta+1}{\theta+2} x^{\theta+2} \Big|_0^1 = \frac{\theta+1}{\theta+2} = \bar{X},$$

解得 α 的矩估计为

$$\alpha = \frac{1-2\bar{X}}{\bar{X}-1}.$$

..... (6分)

(2) 极大似然估计:

$$L(\theta) = \prod_{i=1}^n (\theta+1)x_i^\theta = (\theta+1)^n (x_1 x_2 \cdots x_n)^\theta ,$$

$$\ln L = n \ln(\theta+1) + \theta \sum_{i=1}^n \ln x_i ,$$

$$\frac{d \ln L}{d \theta} = \frac{n}{\theta+1} + \sum_{i=1}^n \ln x_i \stackrel{<}{=} 0 ,$$

解得 α

$$\alpha = -\left(1 + \frac{n}{\sum_{i=1}^n \ln x_i}\right) .$$

因此极大似然估计为 $\alpha = -\left(1 + \frac{n}{\sum_{i=1}^n \ln X_i}\right)$ (6 分)